ML.
- Supervised → Classification / Regression
- Unsupervised
- Reinforcement

Algorithm :- steps.

Linear Regression :-

→ output change proportionaly with input -

House area	Price
1000 ↑	45 ↑
1200	53
1500	65
1700	75
2000	89

positive relationship.

constant rate of change = slope

$y = mx + c$

$y = \beta_0 + \beta_1 x + \varepsilon$ → error.

y → predicted output; β_0 → intercept; β_1 → slope; x → input feature.

predicted price = starting price + (change due to area) + small error.

actual data

area	price
1000	50
1200	60
1500	72

predicted

area	pred.		
1000	48	→ 2	→ 4
1200	62	→ −2	→ 4
1500	69	→ 3	→ 9
			17

Error = Actual − predicted

Residual :→ during training.

RSS :→ Residual Sum of squares.

$RSS = \underbrace{\sum (y_i - \hat{y}_i)}^2$

RSS = 17, RSS = <17

Mean squared error

MSE :-

model A: 100 observation, RSS = 200

model B: 10000 observations, RSS = 200

$$\boxed{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

OLS → Ordinary Least Squares.

Gradient Descent

↓

Direction + speed of change

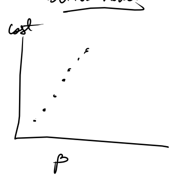
derivative → slope → direction → Gradient.

Actual data → prediction → error → MSE → cost function or loss function

↓

J → smaller score → better model.

	MSE
M_A ⇒	5
M_B ⇒	20



Calculate error
find direction
move a little
check error again
repeat.

Learning rate

- very small lr → very slow learning
- very large lr → overshoots → never converges.
- Good learning rate → fast → stable.

New parameter = Old parameter − (lr × gradient)

iter → 800, cost = 5

iter → 10000, cost → 4.9

Dataset → choose random (β_0 + β_2) → predict $\hat{y}$ → calculate residuals → calculate MSE (cost) → compute gradient → update β_0 + β_2 → repeat → convergence → best fit line.

Data → Random parameters → prediction → calculate error → find gradient → update parameter → Repeat → Model ready.

$y = x^2$, $\frac{dy}{dx} = 2x$

$y = 2$, $\frac{dy}{dx} = 0$